

CSE 4849 / CSE 4885
Human-Computer Interaction (HCI)
Lecture 2: The Human

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The Human

- Human Cognition and Information Processing
- Information i/o ...
 - Visual, Auditory, Touch, Taste, Smell
- Information stored in memory
 - Sensory, Short-term, Long-term
- Information processed and applied
 - Reasoning, problem solving, skill, error
- Emotion influences human capabilities
- Each person is different

Cognitive psychology

- It is the study of human **perception**, **attention**, **memory** and **knowledge**, and the ways in which these have been applied in the design of computer interfaces.
- Cognitive psychology relates the use of computer systems:
 - How humans perceive the world around them
 - How they store and process information to solve problems
 - How they physically manipulate objects

Cognition

- Cognition is the process by which the human gain knowledge.
- The processes which contribute to *cognition* include:
 - Attention
 - Perception and recognition
 - Memory
 - Learning
 - Reading, speaking, and listening
 - Problem-solving, planning, reasoning, decision making
- A key aim of HCI is to understand how humans interact with computers, and to represent how knowledge is passed between the two.

Modes of Cognition

- In 1993, Norman proposed two general mode of human cognition:
 - Experiential cognition: perceive, act, and react. E.g. driving a car, reading a book, having a conversation, playing a video game etc.
 - Reflective cognition: thinking, comparing, and decision making. Designing a car model, authoring a book, designing conversation-based agent, developing a video game, learning.
- Fast (instinctive, reflexive, effortless) Vs Slow thinking (logical thinking, mental effort demanding, concentration)

Perception

Perception

Representations of information need to be designed to be perceptible and recognizable across different media.

- Design icons and other graphical representations so that users can readily distinguish between them.
- Using white space is an effective visual method for grouping information that makes it easier to perceive and locate items.
- Design audio sounds to be readily distinguishable from one another so that users can perceive how they differ and remember what each one represents.
- Research color contrast techniques when designing an interface, especially when choosing a color for text so that it stands out from the background. For example, it is OK to use yellow text on a black or blue background, but not on a white or green background. When choosing colors, also take into account color vision deficiency.
- Haptic feedback should be used judiciously. The kinds of haptics used should be easily distinguishable so that, for example, the sensation of squeezing is represented in a tactile form that is different from the sensation of pushing. Overuse of haptics can cause confusion. ■

Attention

Attention

- Consider context. Make information salient when it requires attention at a given stage of a task.
- Use techniques to achieve this when designing visual interfaces, such as animated graphics, color, underlining, ordering of items, sequencing of different information, and spacing of items.
- Avoid cluttering visual interfaces with too much information. This applies especially to the use of color and graphics: It is tempting to use lots of these attributes, which results in a mishmash of media that is distracting and annoying rather than helping the user attend to relevant information.
- Consider designing different ways of supporting effective switching and returning to a particular interface. This could be done subtly, such as the use of pulsing lights gradually getting brighter, or abruptly, such as the use of alerting sounds or voice. How much competing visual information or ambient sound is present also needs to be considered. ■

Cognitive Architecture

- Cognitive architectures are designed to simulate human intelligence in a humanlike way (Newell, 1990).
 - For example, the chess program that defeated Kasparov, Deep Blue, would not qualify as a cognitive architecture because it does not solve the problem (chess) in a humanlike way.
- Young (Gray, Young, & Kirschenbaum, 1997; Ritter & Young, 2001) defined a cognitive architecture as an embodiment of *“a scientific hypothesis about those aspects of human cognition that are relatively constant over time and relatively independent of task.”*

Cognitive Architecture: Characteristics

- Cognitive architecture is integrative—that is, architectures include mechanisms to support attention, memory, problem solving, decision making, learning, and so forth.
- Cognitive architecture has to be supplied with knowledge needed to perform a particular task.

Cognitive architecture + Related set of knowledge = Computation Cognitive Model

- Cognitive computational model can be considered as quantitative model as they can produce execution times, error rates, and even learning curves.

Model Human Processor (MHP)

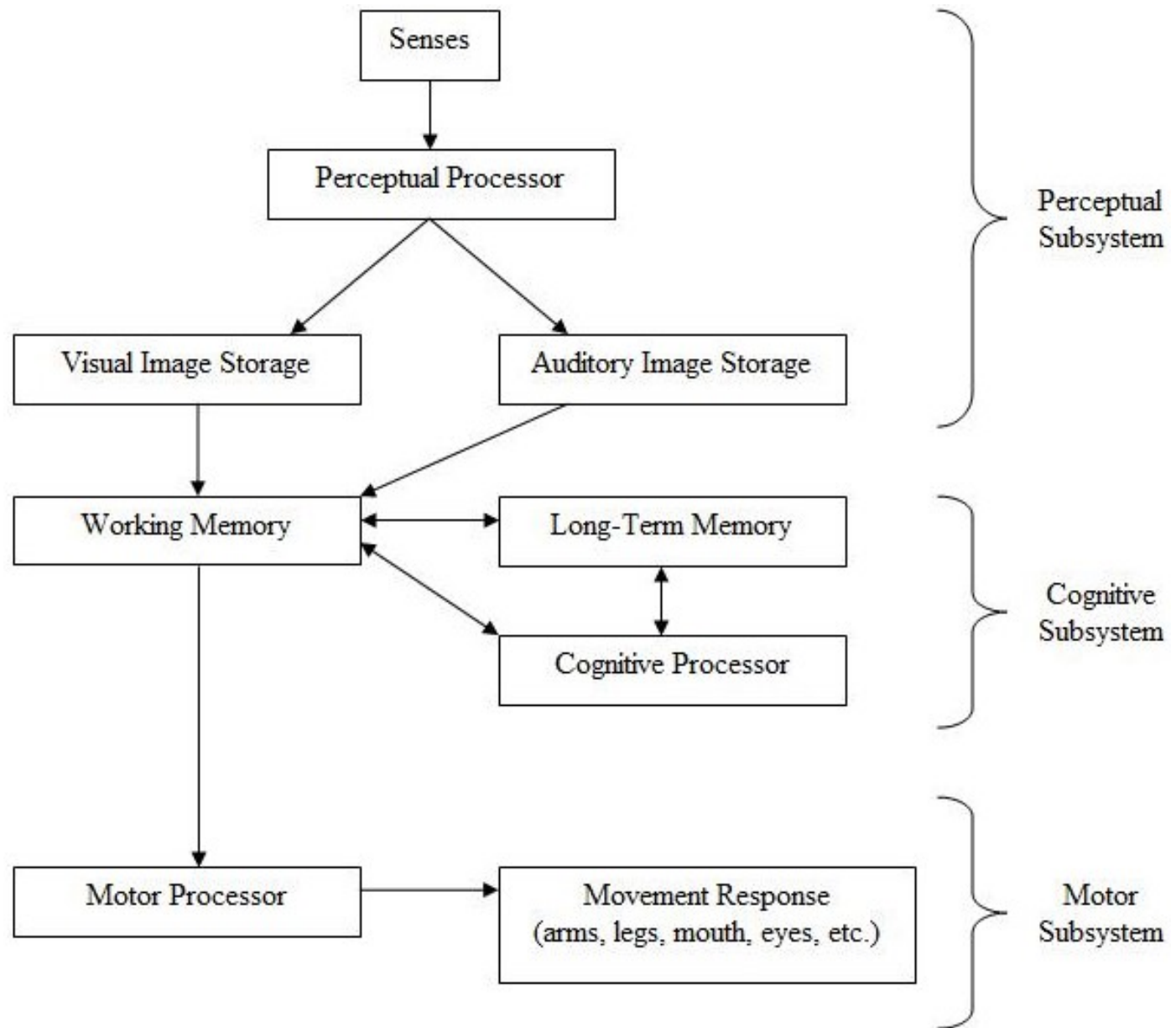
- MHP describes the cognitive process that people go through between perception and action.
- It is important to the study of HCI because cognitive processing can have a significant effect on performance, including task completion time, number of errors, and ease of use.
- This model is based on the simplified view of the human processing involved in interacting with computer systems. (Card, Moran, and Newell, 1983).

Card, S. K., Moran, T. P., & Newell, A. (1983). *The Psychology of Human-Computer Interaction*. Hillsdale, NJ: Lawrence Erlbaum Associates.

Model Human Processor (MHP)...

- The model human processor consists of three interacting systems. Each has its own memory and processor.
- **Perceptual processor:** Handles sensory stimulus from the outside world
 - outputs into audio storage
 - outputs into visual storage
- **Cognitive processor:** Provides the processing needed to connect Perceptual and Motor systems
 - Outputs into working memory
 - has access to:
 - working memory
 - long term memory
- **Motor processor:** Controls actions
 - carries out actions

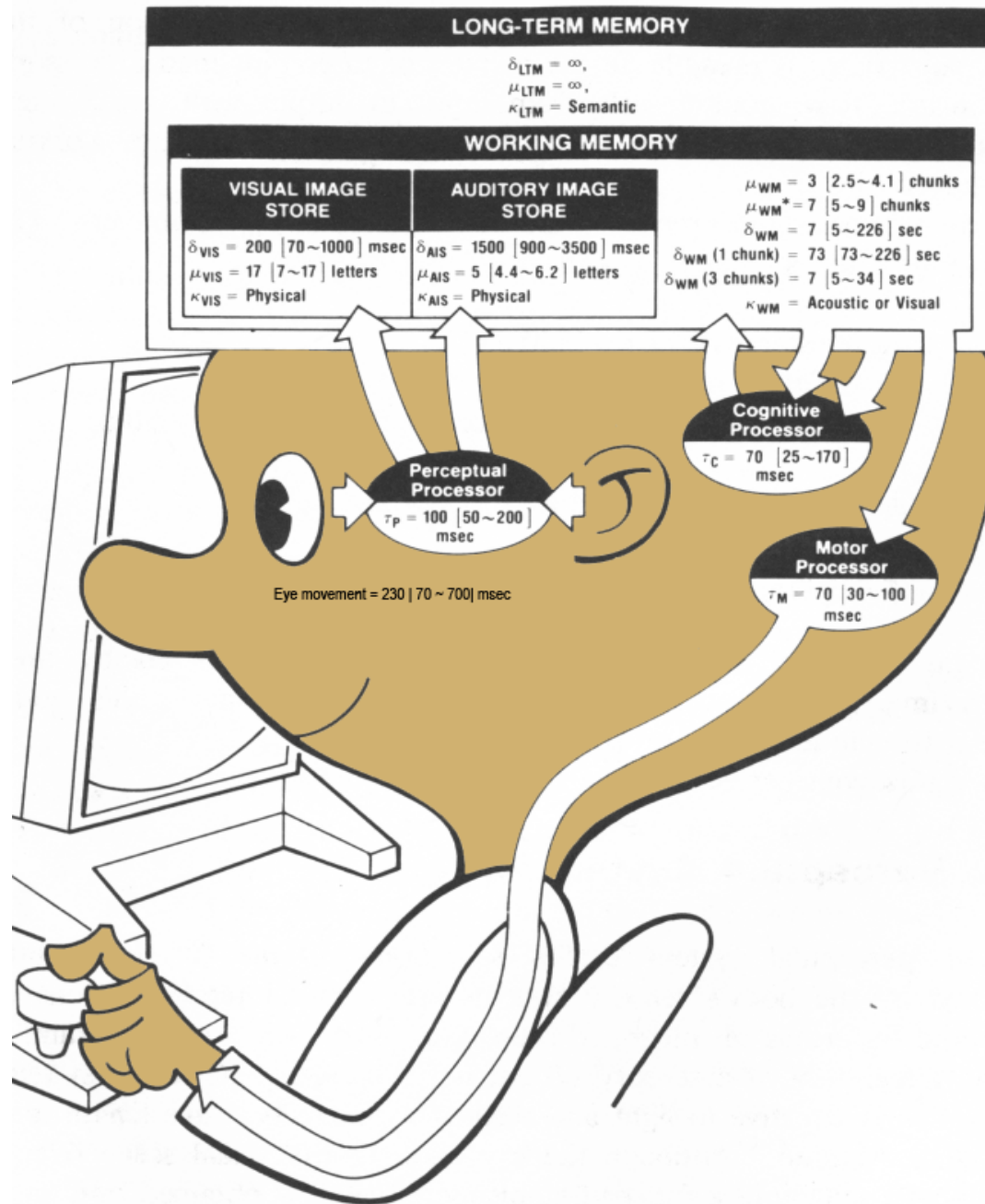
Model Human Processor (MHP)



Why Is the MHP Useful?

- Use empirical studies to validate the model
 - Validates our understanding of the three systems
- Use model to:
 - Predict and compare usability of different interface designs
 - Task performance, learnability, and error rates
 - No users or functional prototype required!
 - Develop guidelines for interface design
 - Color, spatial layout, recall, response rates, etc.
- To be useful, a model must:
 - Be easy to use and learn
 - Produce reasonably accurate results

Model Human Processor (MHP)



- **Assume perceptual cycle time = 100ms**
- If 20 clicks per second are played for 5 seconds, about how many clicks could a person hear?
- If 30 clicks per second are played for 5 seconds, about how many clicks could a person hear?

GOMS Model

- According to the GOMS model, cognitive structure consists of four components:
 - (1) a set of goals,
 - (2) a set of operators,
 - (3) a set of methods for achieving the goals, and
 - (4) a set of selection rules for choosing among competing methods. (IF THEN rules)
- Example: Deleting a sentence (Goal)

m1: Menu method - delete sentence
s1: Highlight sent.
s2: Open menu
s3: Select delete command
s4: Accomplish —
m2: Del-key method

GOMS

- Goals are
 - what a user tries to accomplish, and can be defined at different levels of abstraction.
 - They can also be broken down into sub-goals, thus constructing a hierarchy.
- Operators are
 - Elementary actions which are used to accomplish a goal. They can be perceptual cognitive or motor.
 - They are atomic and cannot be further decomposed. Different GOMS variants have different operators.
 - Typically they are context free, that is, their execution time does not depend on the current state of the system.
- Methods are
 - Procedural algorithms for accomplishing a goal (or subgoal). Several methods may accomplish the same goal.
- Finally, selection rules are
 - used to select the appropriate method for achieving a specific goal. They represent the user's knowledge of how to accomplish the task.

GOMS Example: Move Word (1 / 2)

Goal: move the word starting at the cursor position to the end of the text

[select **use-keyboard**
 delete-and-write
 use-mouse]

verify move

Main goal
with methods

Goal: **use-keyboard**

Goal: select word

[select use <shift> and n^* <cursor right>
 use <shift> and <ctrl> and <cursor right>]

verify selection

...

Sub-goal

Method 1

Goal: **delete-and-write**

...

Method 2

Goal: **use-mouse**

Goal: select word

[select click at beginning and drag till the end of the word
 double-click on the word]

verify selection

Goal: move word

[select click on word and drag till end of text
 Goal: **copy-paste-with-mouse**

...]

Method 3

GOMS Example: Move Word (2 / 2)

- Selection rules:
 - Rule 1: use method **use-keyboard** if no mouse attached
 - Rule 2: use method **delete-and-write** if length of word < 4
 - Rule 3: use method **use-mouse** if hand at mouse before action
 - ...
- Selection rules depend on the user (→ remember user diversity?)
- GOMS models can be derived in various levels of abstraction
 - e.g. goal: write a paper about X
 - e.g. goal: open the print dialog

GOMS Example: Closing a Window

GOAL: CLOSE-WINDOW

[select GOAL: USE-MENU-METHOD

MOVE-MOUSE-TO-FILE-MENU

PULL-DOWN-FILE-MENU

CLICK-OVER-CLOSE-OPTION

GOAL: USE-CTRL-F4-METHOD

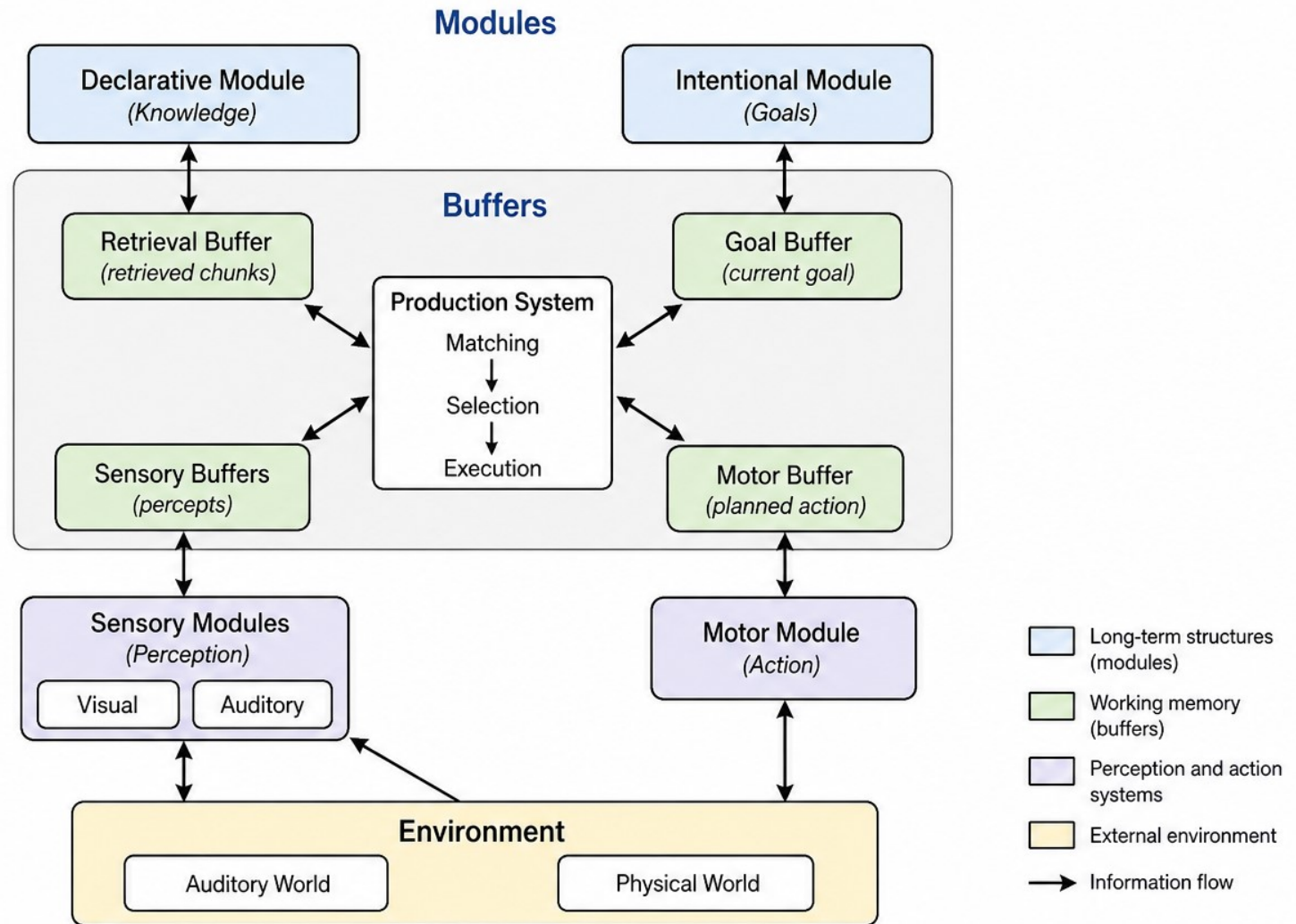
PRESS-CONTROL-F4-KEYS]

For a particular user:

Rule 1: Select USE-MENU-METHOD unless another rule applies

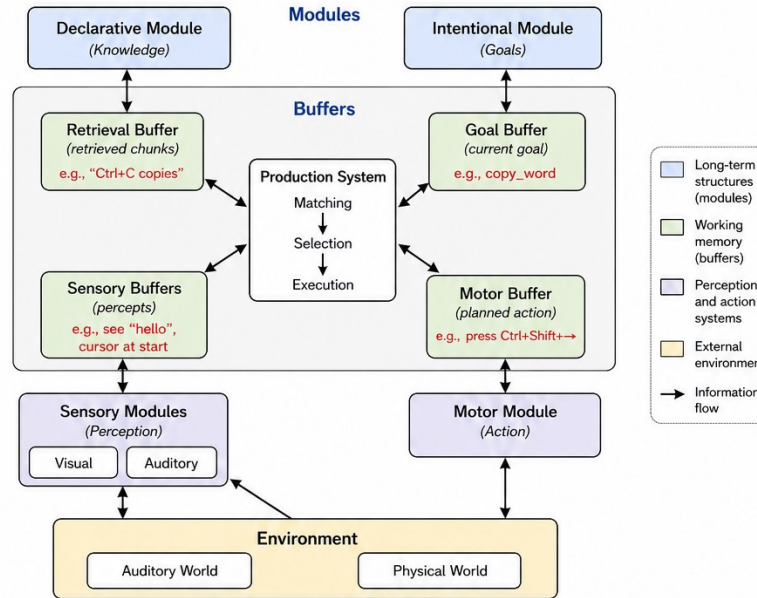
Rule 2: If the application is GAME,
select CTRL-F4-METHOD

Adaptive Control of Thought-Rational (ACT-R)



Source (for correctness): Anderson, J. R. (2007). *How can the human mind occur in the physical universe?* Oxford University Press. (See Chapter 6: "The ACT-R Architecture", pp. 111–134.)

EXAMPLE TASK: Copy the word “hello” and paste it at the end of the sentence using keyboard.

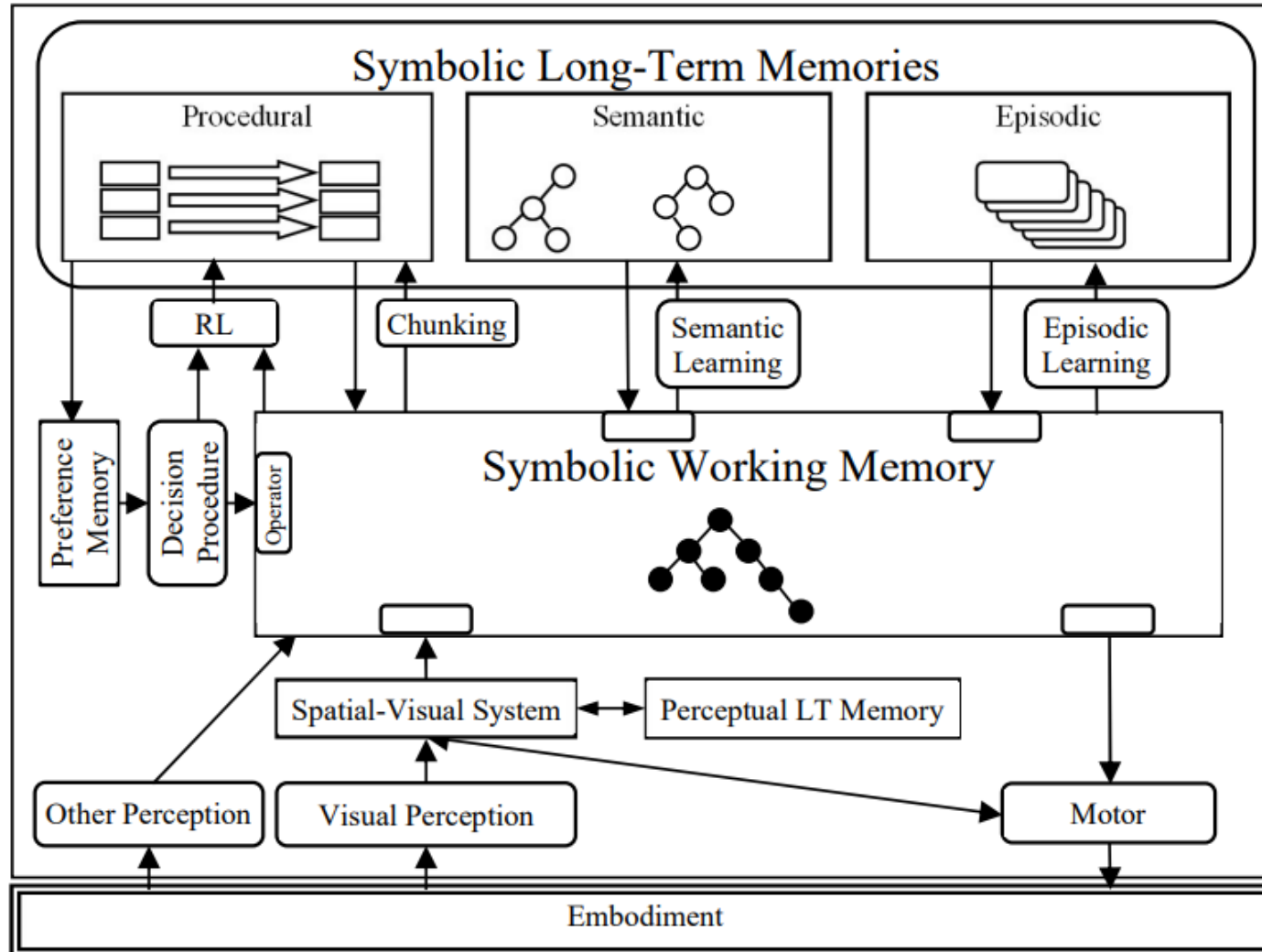


Walkthrough of the Example

Step	What happens	ACT-R components involved	Details
1	See the word "hello"	Environment → Visual module → Sensory buffer	- The word "hello" and the cursor position are perceived. - Sensory buffer holds: "word=hello, cursor_at_start"
2	Goal is active	Intentional module → Goal buffer	- Goal buffer holds: copy_word (Overall task intention: copy then paste at end)
3	Decide what to do	Production system (Matching → Selection → Execution)	- Matches rules with current goal and percepts. - Selects rule: IF goal=copy_word AND cursor_at_word_start THEN select_word_using_keyboard - Executes the rule.
4	Retrieve needed knowledge	Declarative module → Retrieval buffer	- Retrieves chunk: "Ctrl+Shift+→ selects word" - Retrieval buffer now holds this chunk.
5	Prepare action	Production system → Motor buffer	Based on retrieved knowledge, production sets Motor buffer to: press Ctrl+Shift+→
6	Execute action	Motor buffer → Motor module → Environment	- Motor module executes keystrokes (Ctrl+Shift+→). - Word "hello" becomes selected in the physical world.
7	Next action: copy	Production system → Retrieval buffer → Motor buffer → Motor module	- Retrieves "Ctrl+C copies". - Motor buffer: press Ctrl+C - Execute → word is copied to clipboard.
8	Move cursor to end of sentence	Production system → Motor buffer → Motor module	- Plan and execute arrow keys / mouse to move cursor. - Environment updates cursor position.
9	Next action: paste	Production system → Retrieval buffer → Motor buffer → Motor module	- Retrieves "Ctrl+V pastes". - Motor buffer: press Ctrl+V - Execute → word "hello" is pasted at the end.
10	Goal achieved	Intentional module → Goal buffer	- Goal buffer updated: task_complete - System returns to monitoring for next goal.

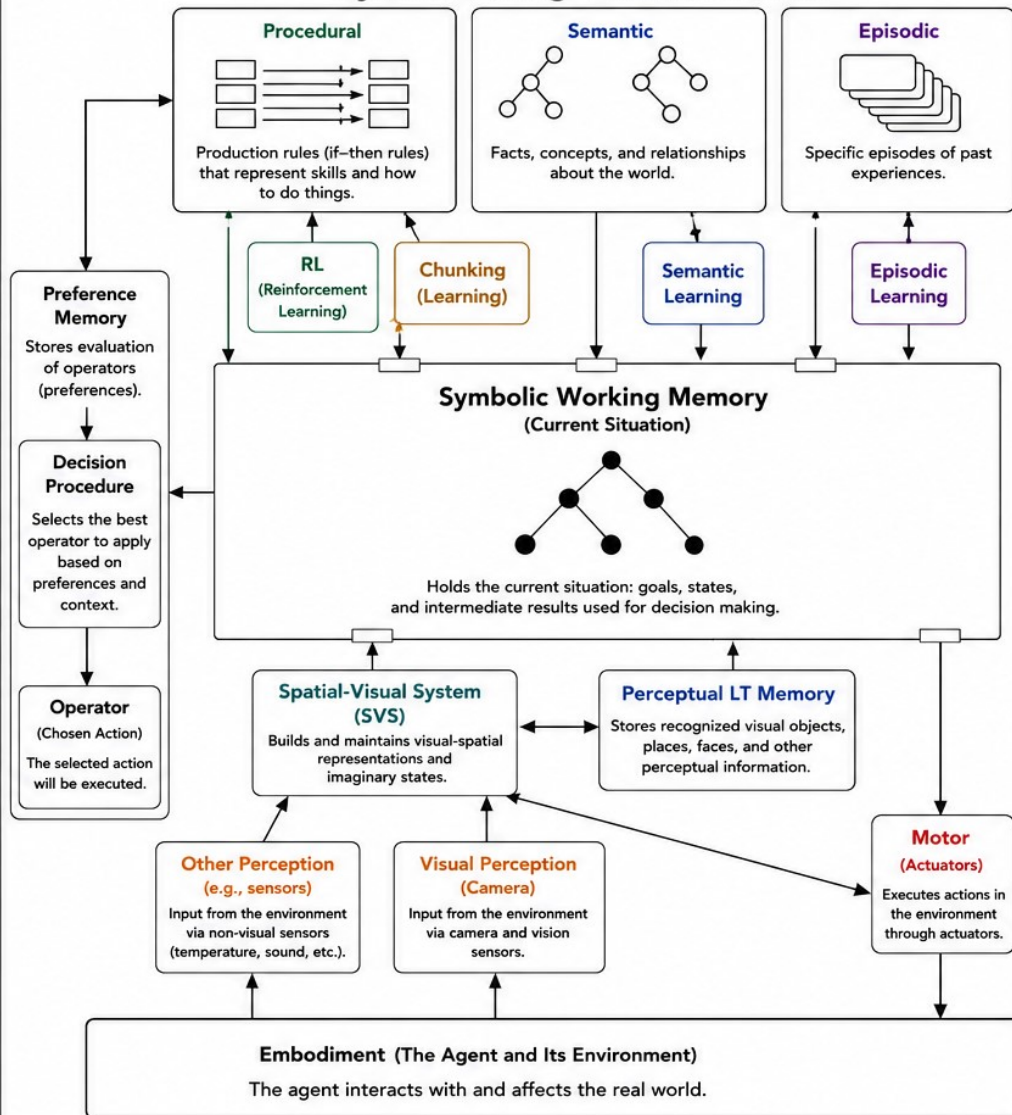
Key idea: The production system controls behavior by cycling through perception → cognition → action using buffers as the interface between long-term modules and the external environment.

SOAR cognitive architecture



<https://arxiv.org/pdf/2205.03854>

Symbolic Long-Term Memories



SOAR ARCHITECTURAL COMPONENTS EXPLAINED

1. Symbolic Long-Term Memories

Store knowledge developed through experience.

- **Procedural Memory:** Production rules (if-then rules) that represent skills and how to do things.
- **Semantic Memory:** Facts, concepts, and relationships about the world.
- **Episodic Memory:** Specific episodes of past experiences.

2. Learning Mechanisms

- **RL (Reinforcement Learning):** Adjusts preferences over operators based on rewards.
- **Chunking (Learning):** Automatically creates new production rules from repeated experiences.
- **Semantic Learning:** Adds or updates facts and concepts.
- **Episodic Learning:** Stores and retrieves episodes.

3. Symbolic Working Memory

Holds the current situation: goals, states, and intermediate results used for decision making.

4. Preference Memory

Stores evaluation of operators (preferences) for decision making.

5. Decision Procedure

Selects the best operator to apply based on preferences and context.

6. Operator (Chosen Action)

The selected operator (action) will be executed.

7. Spatial-Visual System (SVS)

Builds and maintains visual-spatial representations and imaginary states.

8. Perceptual LT Memory

Stores recognized visual objects, places, faces, and other perceptual information.

9. Perceptions

Input from the environment via sensors (visual and other).

10. Motor (Actuators)

Executes actions in the environment through actuators.

11. Embodiment (The Agent and Its Environment)

The agent interacts with and affects the real world.

Cognitive architectures for language agents (CoALA)

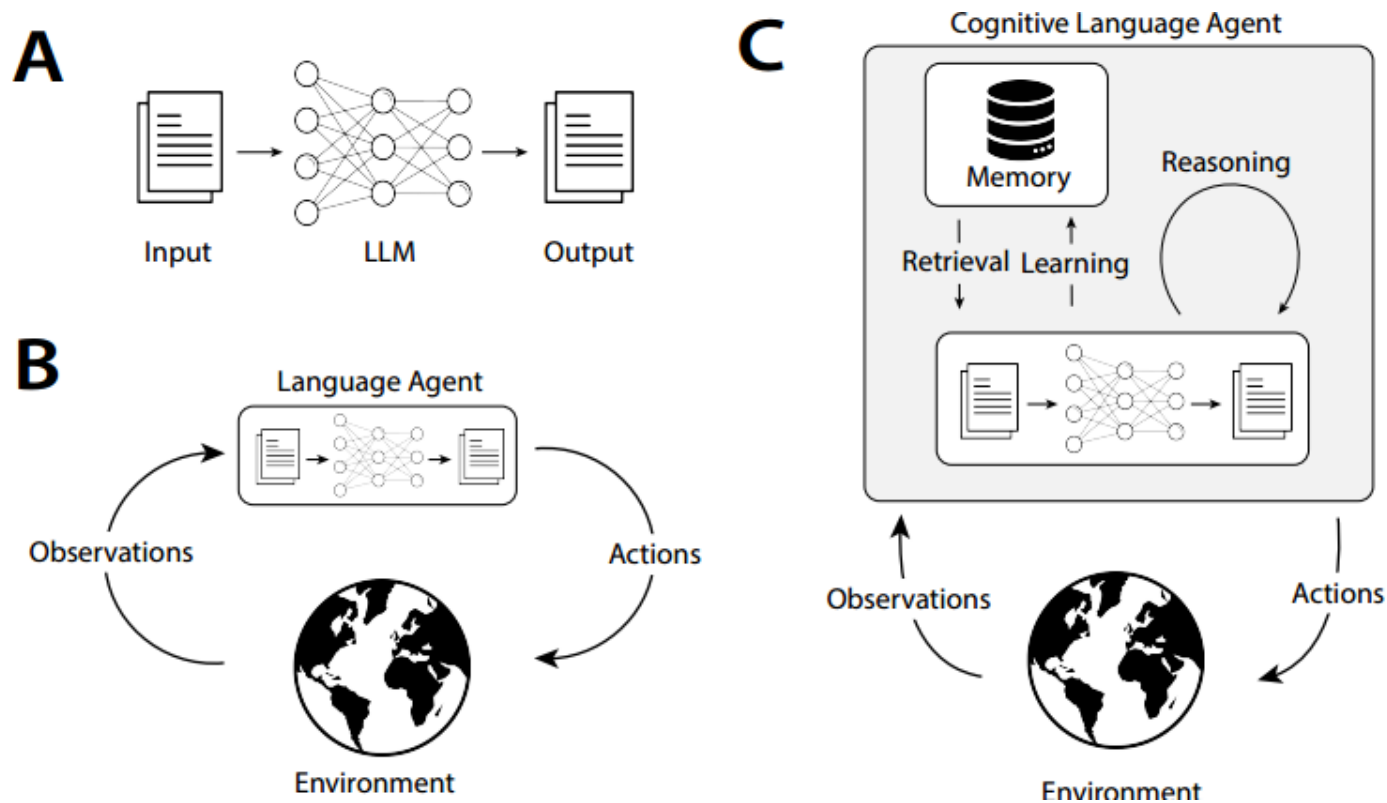


Figure 1: Different uses of large language models (LLMs). **A:** In natural language processing (NLP), an LLM takes text as input and outputs text. **B:** Language agents (Ahn et al., 2022; Huang et al., 2022c) place the LLM in a direct feedback loop with the external environment by transforming observations into text and using the LLM to choose actions. **C:** Cognitive language agents (Yao et al., 2022b; Shinn et al., 2023; Wang et al., 2023a) additionally use the LLM to manage the agent's internal state via processes such as learning and reasoning. In this work, we propose a blueprint to structure such agents.

Cognitive architectures for language agents (CoALA)

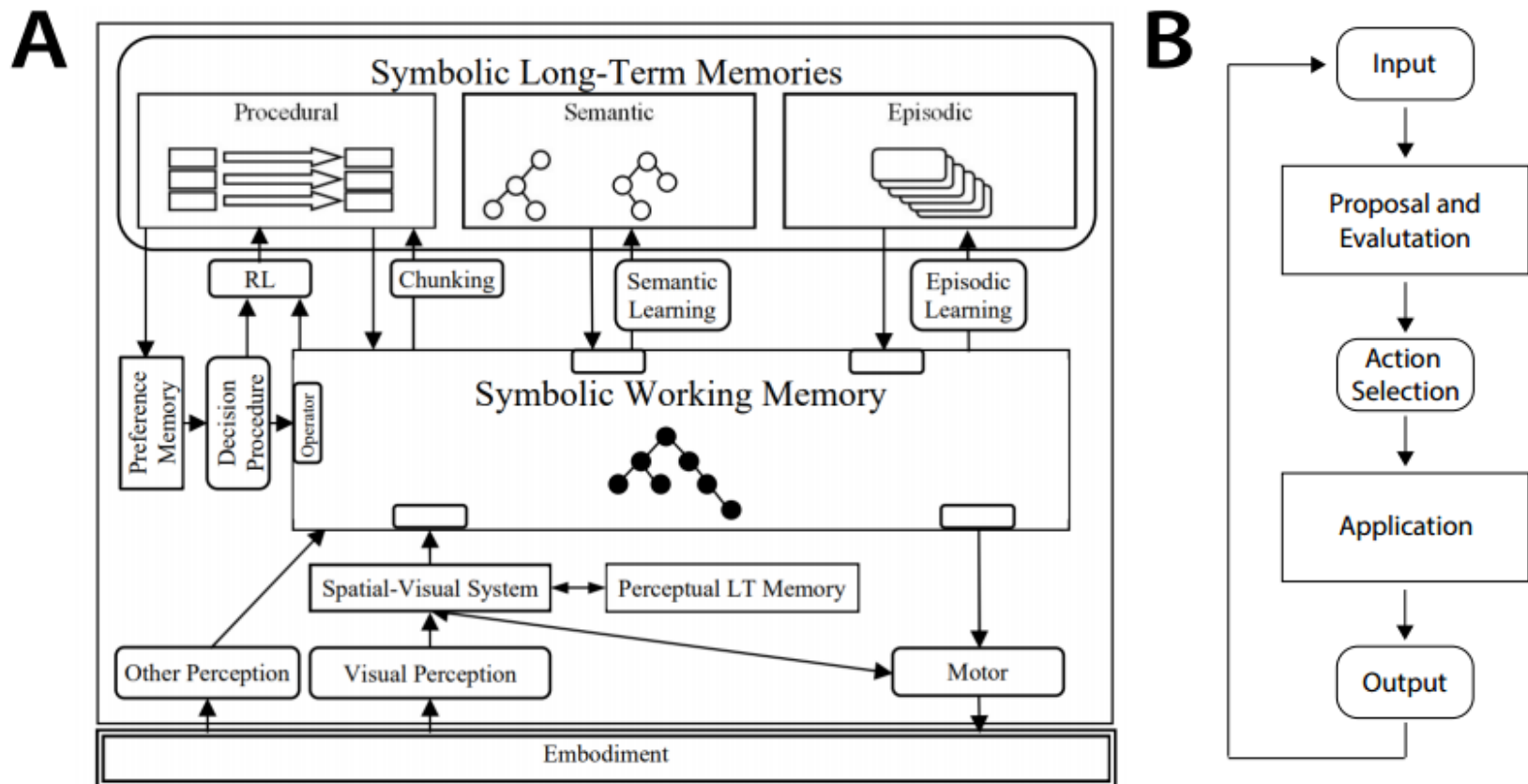


Figure 2: Cognitive architectures augment a production system with sensory groundings, long-term memory, and a decision procedure for selecting actions. **A:** The Soar architecture, reproduced with permission from Laird (2022). **B:** Soar's decision procedure uses productions to select and implement actions. These actions may be internal (such as modifying the agent's memory) or external (such as a motor command).

Cognitive architectures for language agents (CoALA)

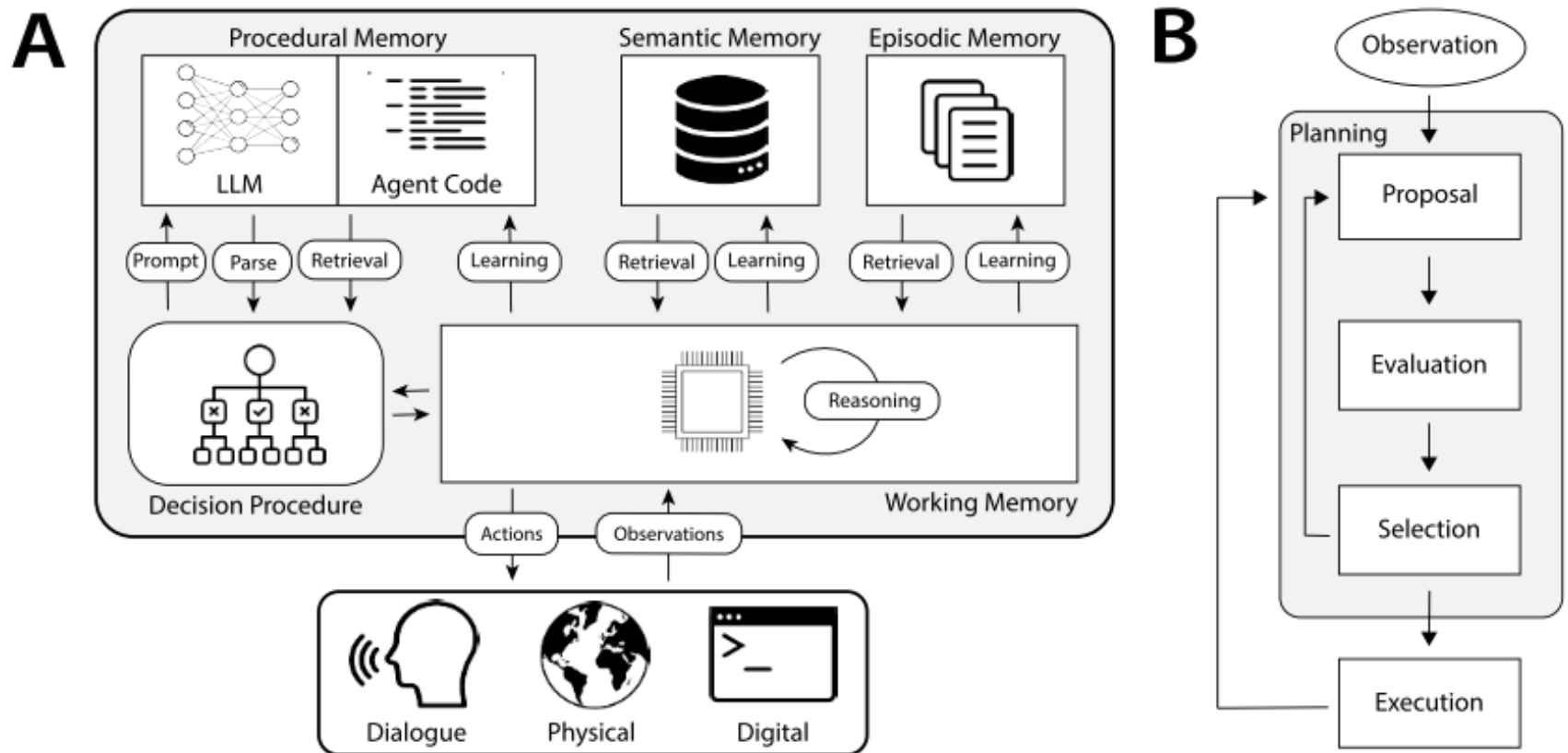


Figure 4: Cognitive architectures for language agents (CoALA). **A:** CoALA defines a set of interacting modules and processes. The **decision procedure** executes the agent's source code. This source code consists of procedures to interact with the LLM (prompt templates and parsers), internal memories (retrieval and learning), and the external environment (grounding). **B:** Temporally, the agent's decision procedure executes a **decision cycle** in a loop with the external environment. During each cycle, the agent uses **retrieval** and **reasoning** to plan by proposing and evaluating candidate **learning** or **grounding** actions. The best action is then selected and executed. An observation may be made, and the cycle begins again.

SOAR Vs CoALA

Diagram Element	SOAR Interpretation	CoALA Interpretation
Procedural Memory	Production rules	LLM + Agent Code
Decision Procedure	Preference-based operator selection	LLM-guided planning
Working Memory	Symbolic structures	Prompt context + scratchpad
Semantic Memory	Symbolic facts	RAG and databases
Episodic Memory	Experience traces	Interaction logs
Reasoning	Rule firing	Chain-of-thought / structured reasoning
Actions	Motor operators	Tool calls and dialogue

Cognitive Theory

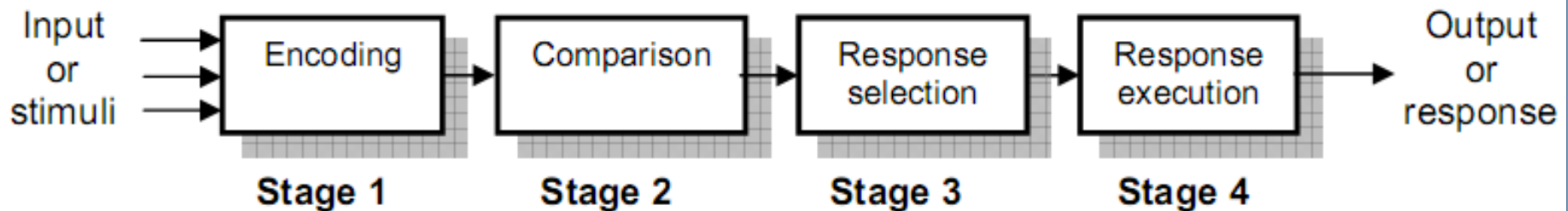
- Human Information Processing
 - *This theory likens the human mind to a computer, suggesting that information is received, processed, stored, and retrieved in a sequential manner.*
- Cognitive Load Theory (CLT)
 - *CLT posits that working memory has limited capacity, and instructional design should avoid overloading it to facilitate learning.*
- Distributed Cognition
 - *Distributed cognition is a theory that suggests cognitive processes are not confined to an individual but are distributed across people, artifacts, and environments.*
- Theory of Mind (ToM)
 - *Theory of Mind refers to the cognitive ability to attribute mental states—such as beliefs, desires, intentions, and emotions—to oneself and others, recognizing that these states may differ among individuals.*

Human Information Processing

- Human performance P , is the function of different information processing stages S ,

$$P = f(S)$$

- The idea of human information processing is that information enters and exits the human mind through a series of ordered stages (Lindsay & Norman, 1977),



Cognitive Load Theory

- CLT emphasizes that human working memory has a limited capacity, which can be overloaded by excessive information or complex tasks.
- It is actually the amount of effort required in reasoning, thinking, and decision making process.
- Three Types of Cognitive Load:
 - Intrinsic Load: This is the load inherent in the task or content itself.
 - Inherent task complexity (regardless of user knowledge)
 - Extrinsic Load: This refers to the load imposed by the way information is presented or the complexity of the user interface.
 - Load added by system design/presentation (can be reduced)
 - Germane Load: This is the load that is beneficial for learning and helps create new knowledge schemas.
 - Mental effort invested during use that promotes learning
- Can you relate this theory with LLM responses?

Example of CLT in LLM

Cognitive Load Type

Relation to LLM

Intrinsic Load

Depends on task complexity (e.g., writing code, learning physics). ChatGPT **adapts explanations** to ease this.

Extrinsic Load

Affected by the way ChatGPT **presents** information — clarity, length, structure.

Germane Load

Encouraged when users use ChatGPT to **deepen understanding**, practice skills, or reflect on responses.

Distributed Cognition

- **Distributed cognition** is the idea that **thinking doesn't happen only inside a person's head**—it is distributed **across people, objects, technology, and the environment** involved in the activity.
 - Cognition is shared across individuals (collaborative work).
 - Artifacts/tools (like checklists, maps, software interfaces) carry part of the cognitive load.
 - Environment affects and supports cognitive processes (e.g., layout of a control room).
 - Cognition happens in interactions, not just in internal mental processing.

Theory of Mind

- **Theory of Mind (ToM)**

- Theory of Mind refers to the cognitive ability to attribute mental states—such as beliefs, desires, intentions, and emotions—to oneself and others, recognizing that these states may differ among individuals.

- **HCI Application:**

Incorporating ToM principles can enhance user experience by designing systems that anticipate user needs and adapt accordingly.

- **Practical Example:**

You're in a group project, and a teammate misses a meeting. You think, “Maybe they didn’t know the time was changed” — you’re attributing a belief (or lack of knowledge) to that person.

Theory of Mind

LLMs Moving Toward ToM

Recent research tries to enhance ToM-like capabilities in LLMs through:

- **Memory-enhanced agents** (tracking beliefs or intents across sessions),
- **Meta-cognition modeling** (e.g., reasoning about uncertainty),
- **Multi-agent systems** (where agents model each other's beliefs or goals).

Tutorial

- Suppose you want to design a file searching interface for smart phones and optimize two types memory processes, one is memory recall and other one is memory recognition.
- Your task is to design a file searching output interface keeping in mind the following concepts:
 - Attention and sliced attention
 - Memory / cognitive load
 - Response time
 - Cognitive computation model
 - Human information processing steps

Tutorial

- Imagine building a **therapy assistant chatbot** for depression or anxiety.
 - Identify and design each CoALA module in their system.
 - Create an **interaction flowchart** showing how the chatbot would process, remember, adapt, and act during a single therapy session.
 - Discuss **ethical implications** (e.g., safety, bias, emotional manipulation).

Tutorial

- Suppose you want to design an information based medical health care app. The target users are mostly rural people from remote area. Your app will take query in natural text writing/a set of selection menus/query dialogue and give relevant information to the question. The challenge is:
 - How to display those health help information to those users in terms of human cognition with an aim to minimize cognitive load
 - If you introduce password protection how they will remember or memorize the password.
 - How to feel them ease and comfortable while using the app.

Input-Output Channels

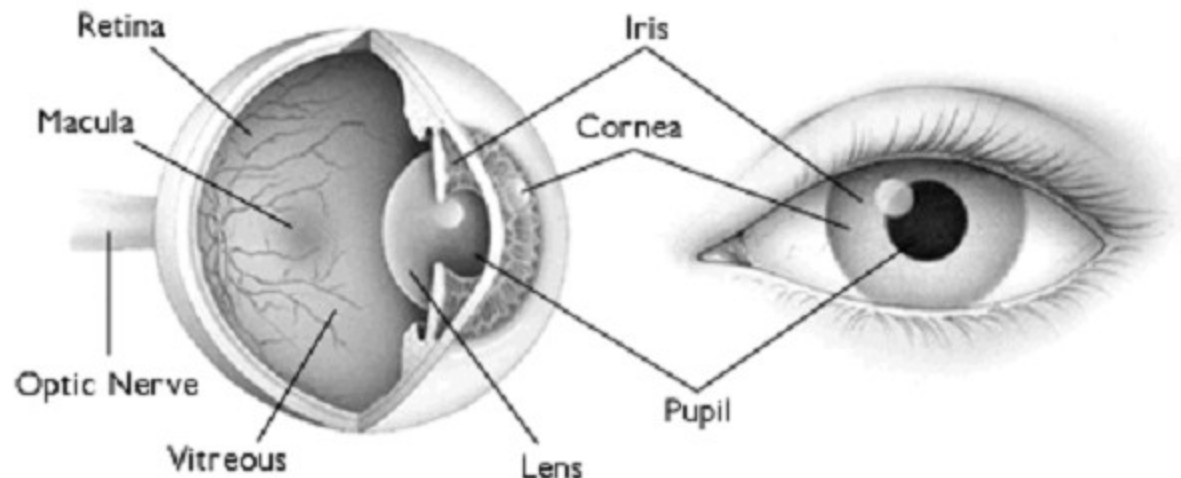
- Information i/o ...
 - Vision
 - Hearing
 - Touch

Vision

- Human vision is a highly complex activity with a range of physical and perceptual limitations, yet it is the primary source of information for the average person.
- Visual perception can be divided in to two stages:
 - Physical reception of stimulus from outside world
 - Processing and interpretation of stimulus

The Eye - physical reception

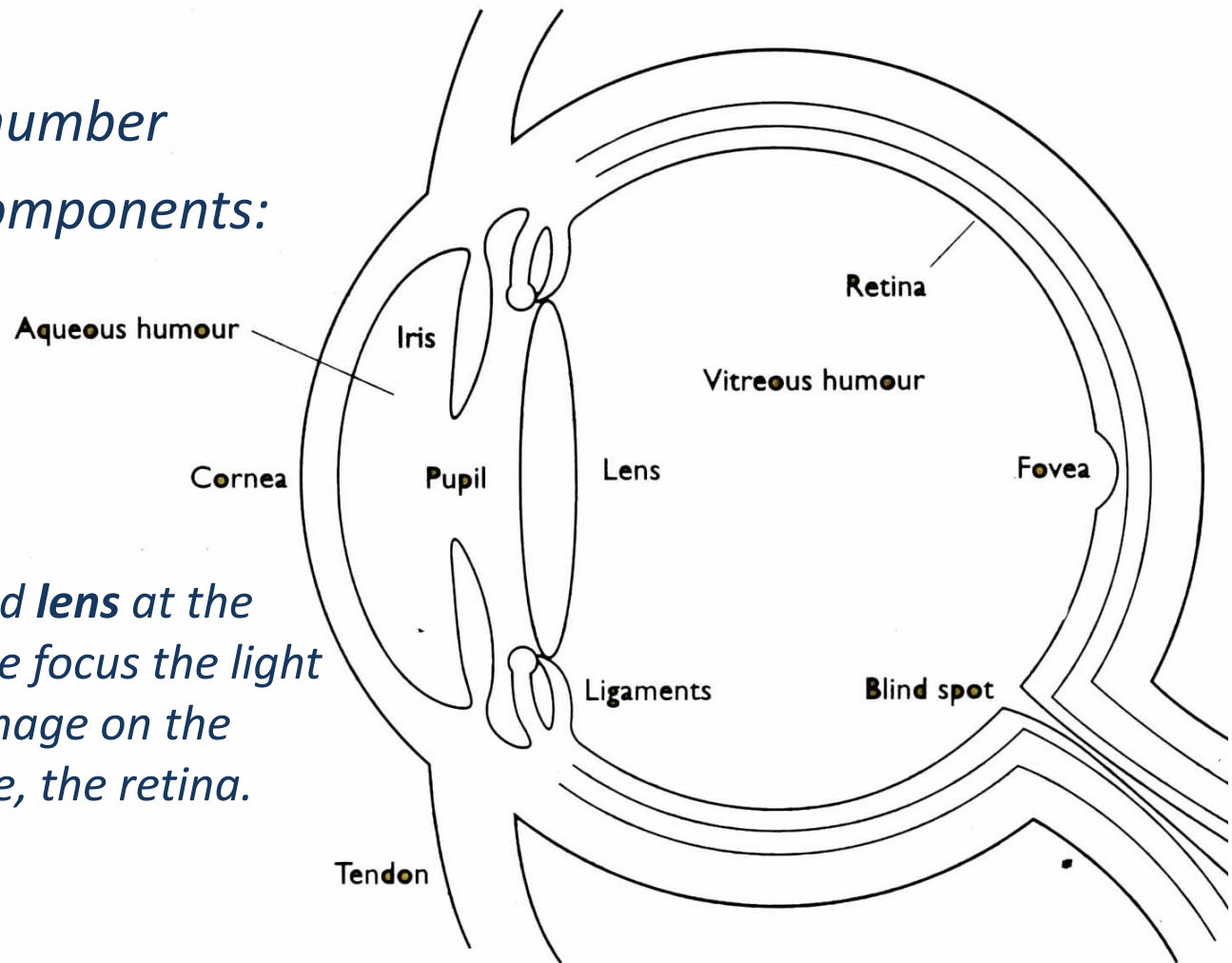
- The eye is a mechanism for receiving light and transforming it into electrical energy
 - Light reflects from objects in the world and their images are focused upside-down on the back of the eye.
 - The receptors in the eye transform it into electrical signals which are passed to the brain.



The Eye - physical reception...

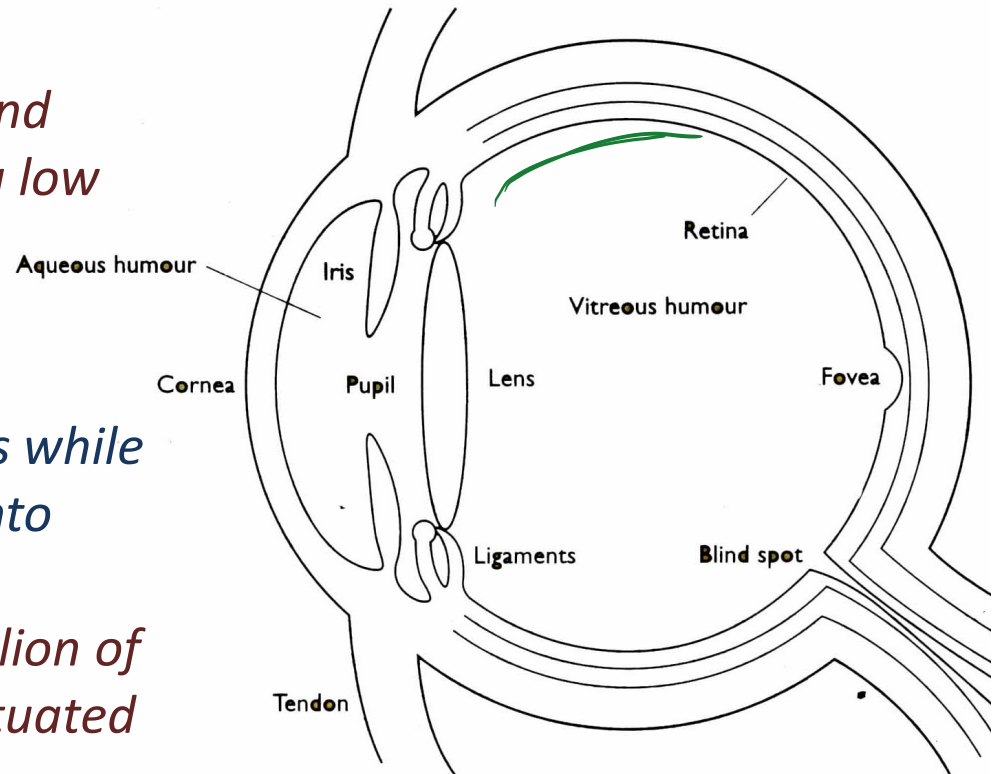
- *The eye has a number of important components:*

- *The **cornea** and **lens** at the front of the eye focus the light into a sharp image on the back of the eye, the retina.*



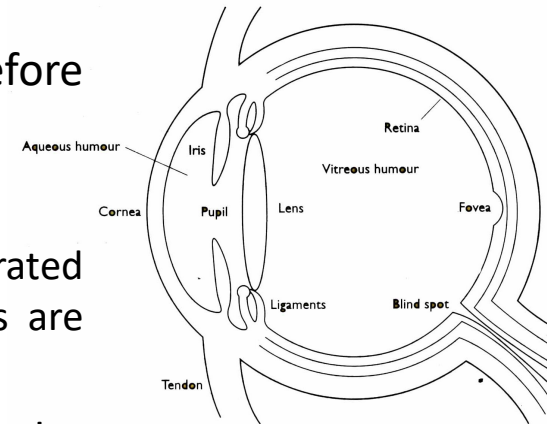
The Eye - physical reception...

- *The retina is light sensitive and contains two types of photoreceptor: rods and cones.*
- *Rods are high sensitive to light and therefore allow us to see under a low level of illumination.*
- *Unable to resolve fine detail and subject to light saturation*
– *reason for temporary blindness while moving from a darkened room into sunlight*
- *There are approximately 120 million of rods per eye which are mainly situated towards the edge of the retina.*
- *Rods dominates the peripheral vision*



The Eye - physical reception...

- **Cones** are less sensitive to light than rods and can therefore tolerate more light.
 - Cones are responsible for colour vision.
 - The eye has approximately 6 million cones, mainly concentrated on the **fovea**, a small area of the retina on which images are fixated
- Retina contains a **blind spot** where the optic nerve enters the eye. The blind spot has no rods and cones.
- The retina has also some specialized cells called ganglion cells. There are two types of **ganglion cells**:
 - **X-cells**: concentrated on fovea and are responsible for the early detection of pattern
 - **Y-cells**: are more widely distributed in the retina and are responsible for the early detection of movement.
 - The distribution of these cells means that, while we may not be able to detect changes in pattern in peripheral vision, we can perceive movement.



The Eye - physical reception...

- Blind spot testing



The Eye - physical reception...

DESIGN FOCUS



Getting noticed

The extensive knowledge about the human visual system can be brought to bear in practical design. For example, our ability to read or distinguish falls off inversely as the distance from our point of focus increases. This is due to the fact that the cones are packed more densely towards the center of our visual field. You can see this in the following image. Fixate on the dot in the center. The letters on the left should all be equally readable, those on the right all equally harder.

A B C D E F • H I J K

This loss of discrimination sets limits on the amount that can be seen or read without moving one's eyes. A user concentrating on the middle of the screen cannot be expected to read help text on the bottom line.

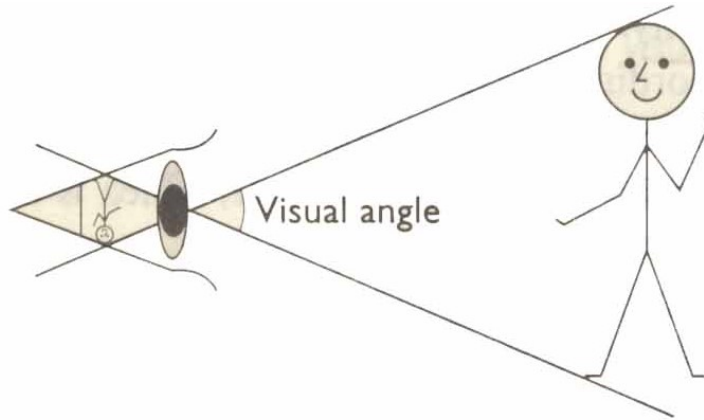
However, although our ability to discriminate static text diminishes, the rods, which are concentrated more in the outer parts of our visual field, are very sensitive to changes; hence we see movement well at the edge of our vision. So if you want a user to see an error message at the bottom of the screen it had better be flashing! On the other hand clever moving icons, however impressive they are, will be distracting even when the user is not looking directly at them.

A B C D E F • H I J K

Interpreting the signal

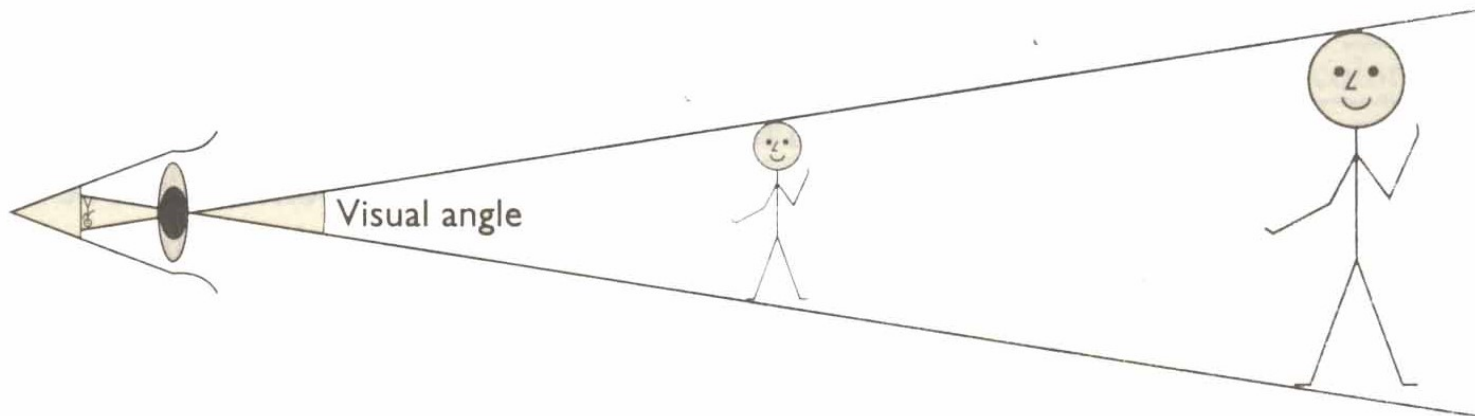
- Perceiving Size
 - The size of image is specified as a visual angle.
 - If we draw a line from the top of the object to central point on the front of the eye and a second line from the bottom of the object to the same point, then the visual angle of the object is the angle between these two lines.
 - visual angle indicates how much of the field of view is taken by the object
 - Visual angle measurement is given by either degree or minutes of arc, where 1 degree equivalent to 60 minutes of arc.

Visual perception – visual angle



Objects of the same size at different distances have different visual angles

Objects of different sizes and different distances may have the same visual angle



Visual angle is affected by both the size of the object and the distance from the eye

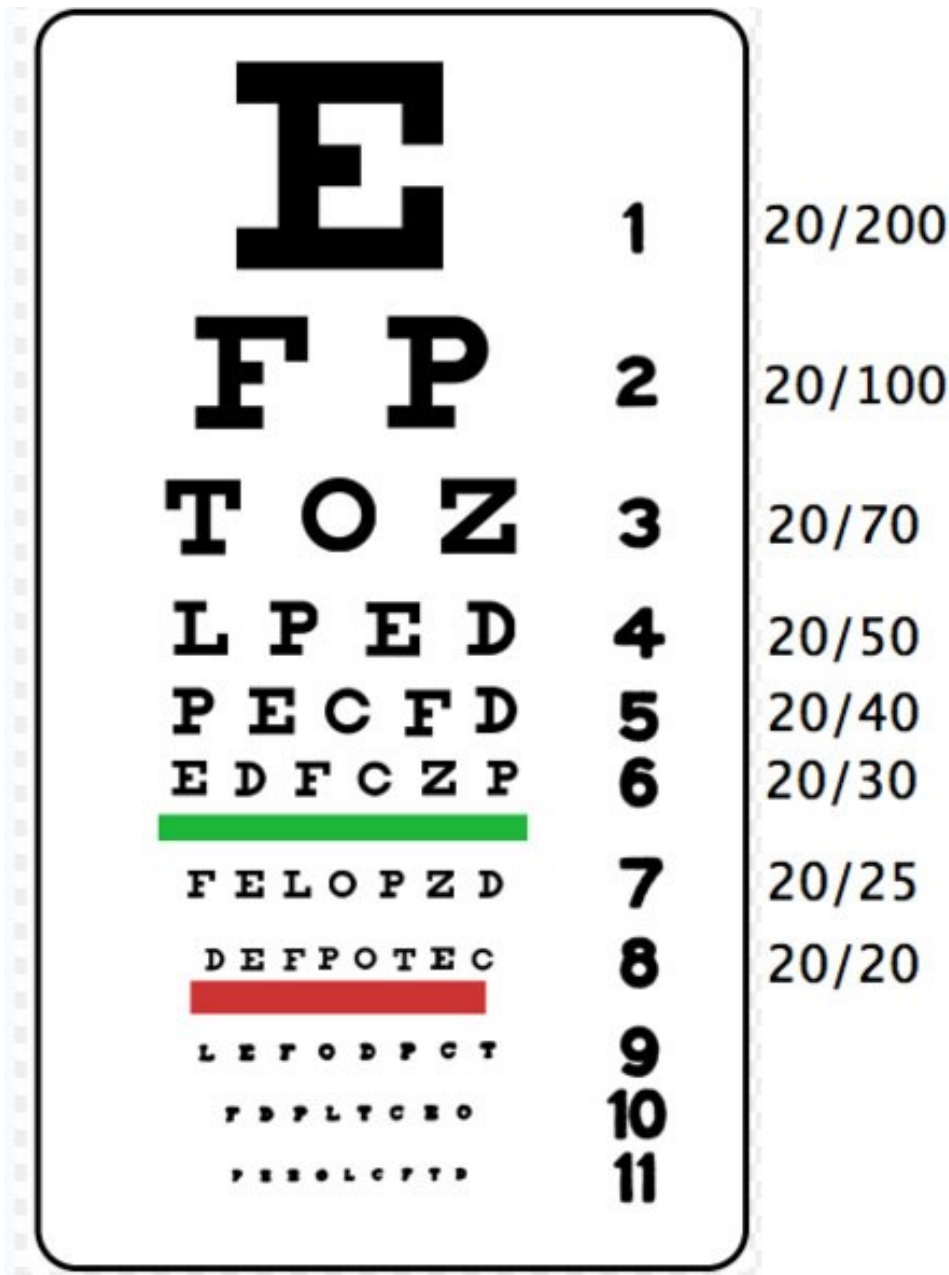
Relationship between Visual Angle, Object Size, Focal Length.

$$\checkmark \quad \text{Visual Angle} = 2 \cdot \tan^{-1} \frac{\frac{\text{Object Size}}{2}}{\text{Object Distance}}$$

$$\checkmark \quad \text{Pixel density} = \frac{\sqrt{\text{width}^2 + \text{length}^2}}{\text{screen_size (diagonal)}}$$

Visual perception – Visual acuity

- Visual acuity is the ability to perceive fine detail (limited)
- familiar objects perceived as constant size
(in spite of changes in visual angle when far away)
- Cues like overlapping help perception of size and depth



Visual perception – Depth

- Human visual system perceives depth using both physiological and psychological cues
- Depth cues can be based on binocular (two eyes) or monocular (one eye) vision
- Physiological cues
 - Accommodation
 - Convergence
 - Binocular parallax
 - Monocular movement parallax
- Psychological cues
 - Retinal image size
 - Linear perspective
 - Texture gradient
 - Overlapping
 - Aerial perspective
 - Shades and shadow

Oculomotor cues

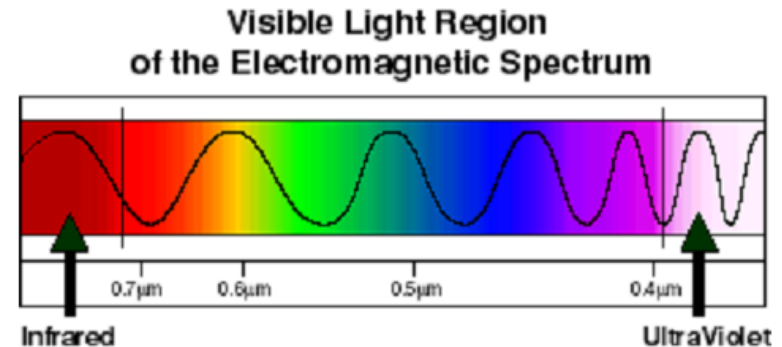
Binocular depth cues

Monocular depth cues



Interpreting the signal...

- Brightness
 - Subjective reaction to levels of light
 - Affected by luminance of object
 - Measured by just noticeable difference
 - Contrast is the function of the luminance of an object and the luminance of its background
 - Visual acuity increases with luminance as does flicker
- Colour
 - Made up of hue, intensity, saturation
 - Cones sensitive to colour wavelengths
 - Blue acuity is lowest
 - 8% males and 1% females colour blind



Perceiving color

- Color is emotional
 - Psychological and cultural
 - Physiological: some colors excite neurons in the brain more than others (red)
- Color is good for segmentation (grouping)
 - Reduce the amount of time to search for information
 - Too many colors can increase search time
- Color is good for drawing attention

Perceiving Color...

- As we age,
 - Lens yellows & absorbs shorter wavelengths that is sensitivity to blue is even more reduced
 - Fluid between lens and retina absorbs more light
 - Perceive a lower level of brightness
- Design implications
 - Do not rely on blue for text or small objects
 - Older users need brighter colors
 - Do not display simultaneously highly saturated extreme colors. i.e. no cyan/blue at the same time as red because it requires constant refocusing.
 - Use pastel colors
 - Opponent color go well. i.e. red and green, yellow and blue etc.
 - Older users need higher brightness levels to distinguish colors.
 - Avoid red and green in the periphery, yellow and blue works in the periphery.
 - Avoid pure blue for text, lines and small shapes.

Interpreting the signal (cont)

- Visual processing involves the transformation and interpretation of a complete image, from the light that is thrown on the retina.
- Our expectations affect the way an image perceived. For example distance is irrelevant to size perception.
- The visual system compensates for:
 - movement
 - changes in luminance.
- Context is used to resolve ambiguity
- Optical illusions sometimes occur due to over compensation

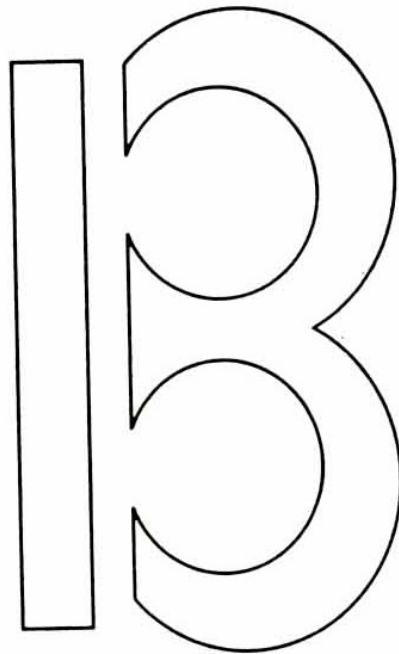


Figure 1.3 An ambiguous shape?

ABC

Figure 1.4 ABC

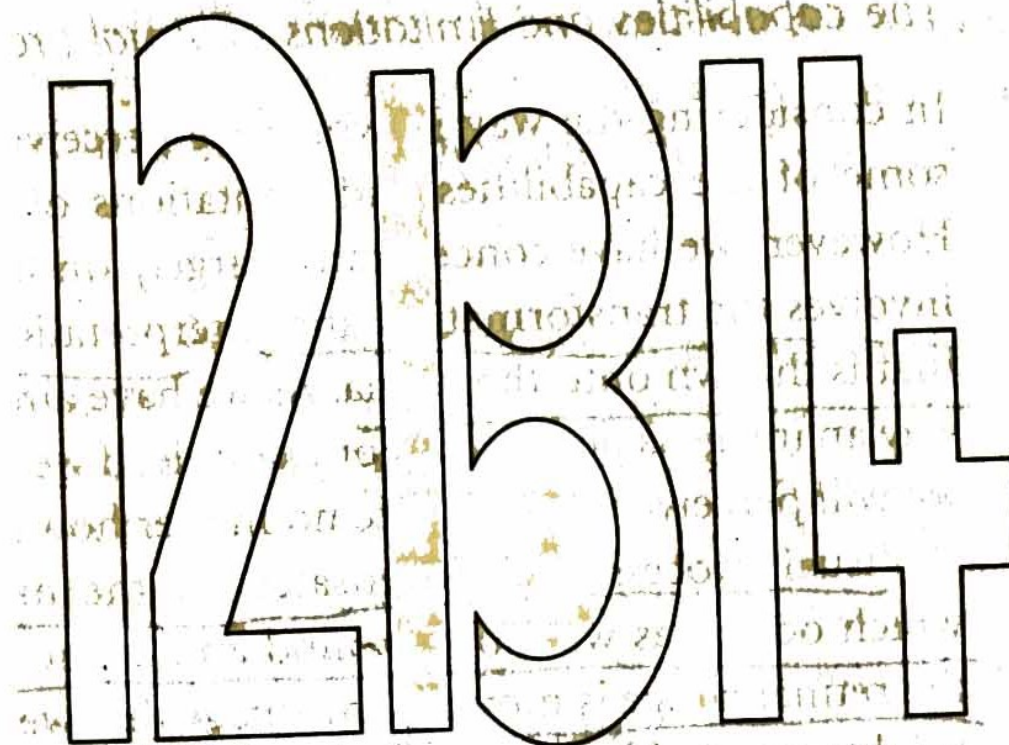


Figure 1.5 1234

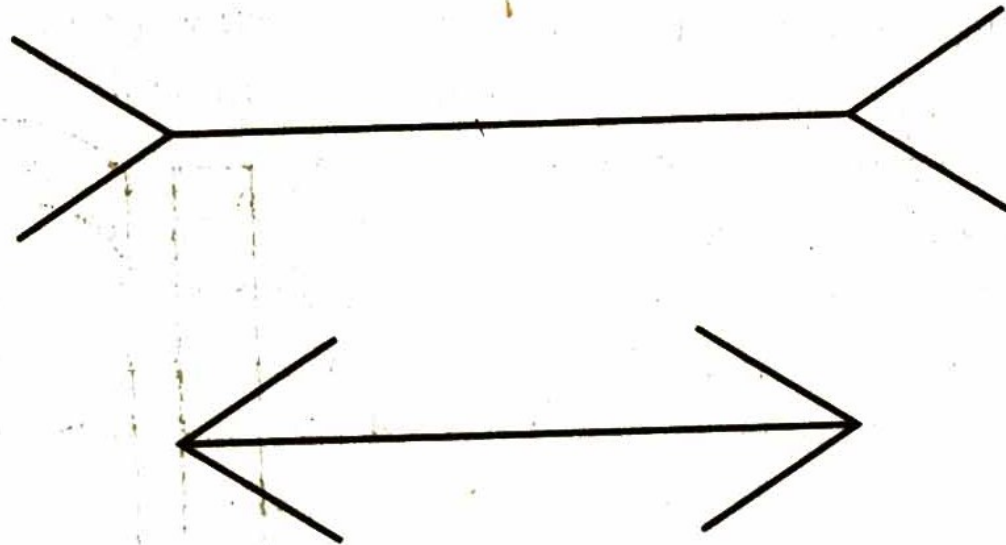


Figure 1.6 The Muller-Lyer illusion – which line is longer?

Optical Illusions

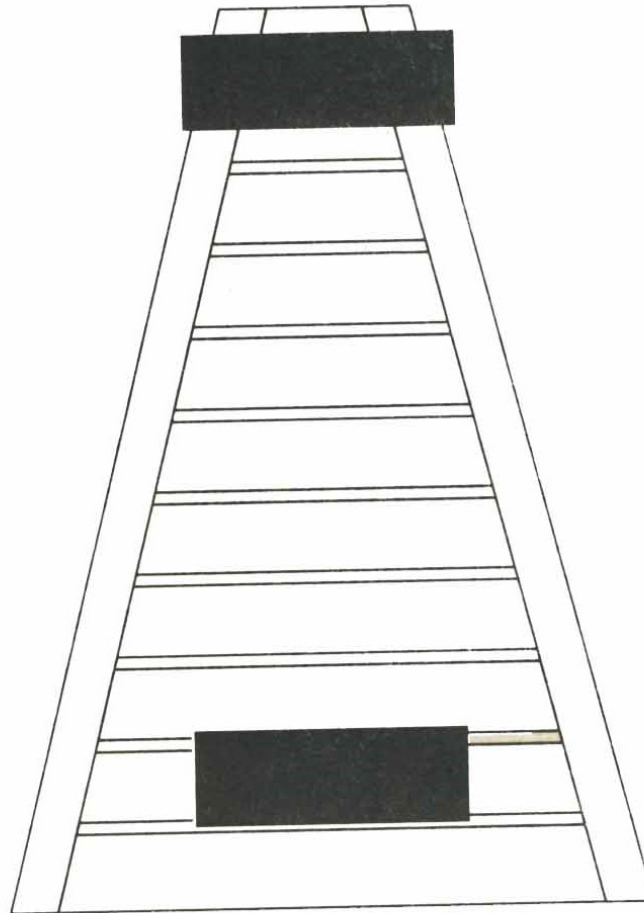
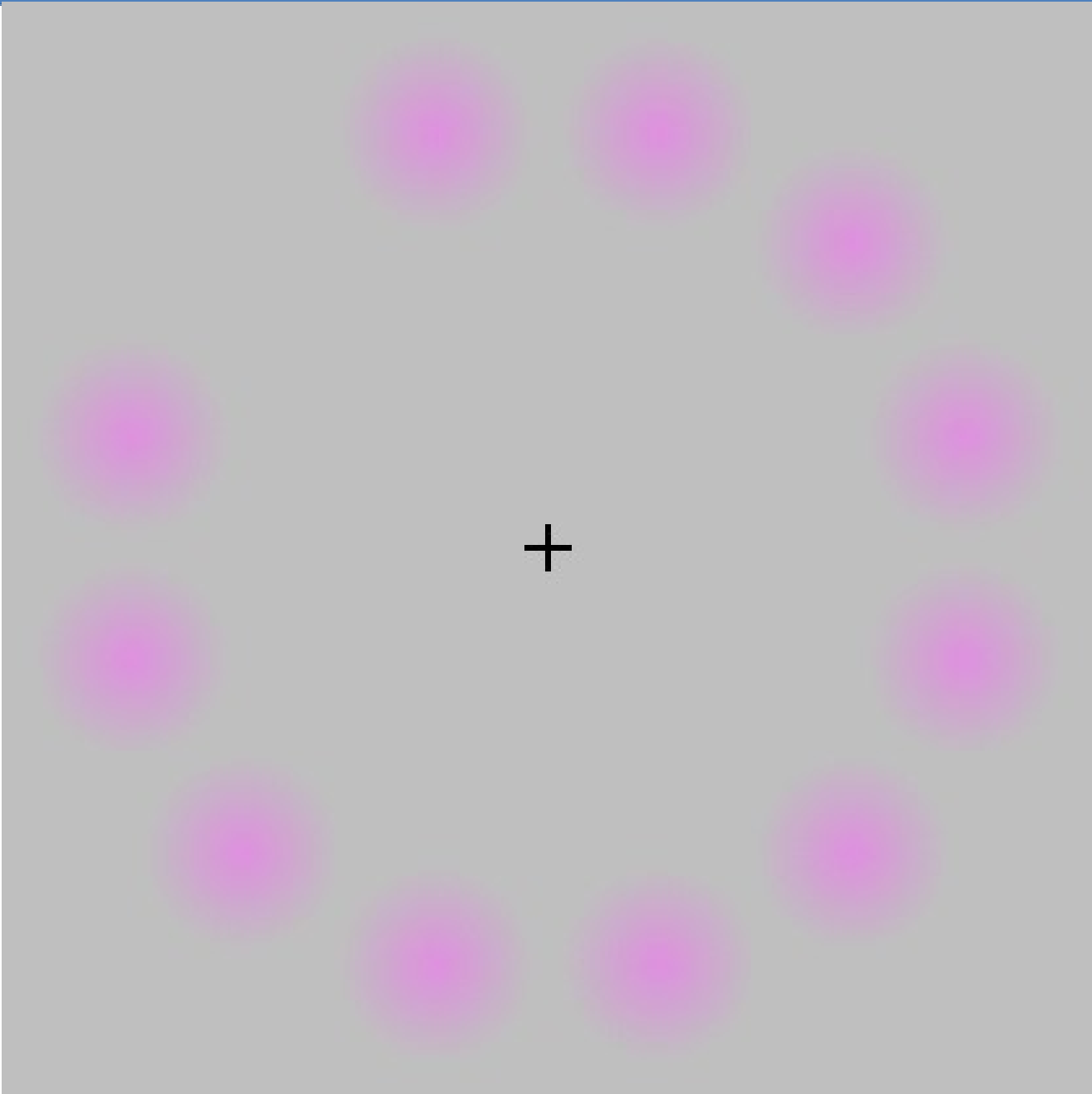


Figure 1.7 The Ponzo illusion – are these the same size?

Optical Illusions

The quick brown
fox jumps over the
the lazy dog.

Figure 1.8 Is this text correct?



Reading Text

- Perception and processing of text is a special case that is important for designing interface.
- The reading process consists of several stages:
 1. The visual pattern of the word on the page is perceived
 2. It is then decoded using internal representation of language
 3. interpreted using knowledge of syntax, semantics, pragmatics
- Reading involves *saccades* and *fixations*
- Perception occurs during fixations periods, which account for approximately 94% of the time elapsed.
- The eye moves backwards over the text as well as forwards, in what are known as regressions. If the text is complex there will be more regressions.

Reading Text...

- The shapes of the first and last letters might be more significant than the shapes of the internal letters

Reading Text...

- Adults read approximately 250 words a minute.
 - It is unlikely that, words are scanned serially, character by character, since experiments have shown that words can be recognized as quickly as single characters
- We read extended text passages more quickly in lowercase than uppercase

WORD SHAPE word shape

- Word shape is important to recognition
 - Removing the word shape clues is detrimental to reading speed and accuracy

Reading Text...

- The speed at which text can be read is a measure of its legibility.
 - Experiments have shown that standard font sizes of 9 to 12 are equally legible.
 - There is a evidence that reading from a computer screen is slower than from a book.
- Negative contrast (dark characters on a light screen) improves reading from computer screen than positive contrast.

Text: design considerations

- Legibility
- Size
- Typefaces
- Type weight
- Kerning
- Leading
- Use of capital letters

Hearing

- The sense of hearing is often considered as secondary to sight.
 - What sounds can you hear?
 - Where are they coming from?
 - What is making them?
 - How far away the sound is?
 - Which direction the sound is coming From ?
- Provides information about environment:
distances, directions, objects etc.

Hearing...

- Physical apparatus:
 - outer ear: protects inner and amplifies sound
 - middle ear: transmits sound waves as vibrations to inner ear
 - inner ear: chemical transmitters are released and cause impulses in auditory nerve
- Sound
 - pitch
 - 20Hz – 15KHz frequency
 - Human is less accurate in distinguishing high frequencies than low ones
 - Tuned to 3KHz by shape of outer ear
 - Loudness
 - Amplitude, is a psychological property of sound
 - Our ears are capable to cope with 0 to 160db (pain at 130db!)
 - Prolonged exposure above 85 dB can cause hearing damage noise-induced hearing loss (NIHL)
 - Timbre
 - type or quality
 - 'signature' of sound source
 - complex set of resonance overlaying the fundamental frequency

Hearing (cont)

- Humans can hear frequencies from 20Hz to 15kHz
 - less accurate distinguishing high frequencies than low.
- Auditory system filters sounds
 - can attend to sounds over background noise.
 - for example, the cocktail party phenomenon.

Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
 - thermoreceptors – heat and cold
 - nociceptors – pain
 - mechanoreceptors – pressure

(some instant, some continuous)
- Some areas more sensitive than others e.g. fingers. (Two point threshold test)
- Kinesthesia - awareness of body position
 - affects comfort and performance.



Movement

- Time taken to respond to stimulus:
reaction time + movement time
- Movement time dependent on age, fitness etc.
- Reaction time - dependent on stimulus type:
 - visual ~ 200ms
 - auditory ~ 150 ms
 - pain ~ 700ms
- Increasing reaction time decreases accuracy in the unskilled operator but not in the skilled operator.

Fitts' Law

Movement ...

Amplitude
width

- Fitts' Law predicts that the time to point at an object using a device is a function of the distance from the target object and the object's size.

- Fitts' Law predicts the time taken in pointing tasks to hit/reach a specific screen target:

$$Mt = a + b \log_2 \left(\frac{D}{S} + 1 \right)$$

where: a and b are empirically determined constants
Mt is movement time
D is Distance
S is Size of target

⇒ *targets as large as possible distances as small as possible*

$\log \frac{A}{w}$

none

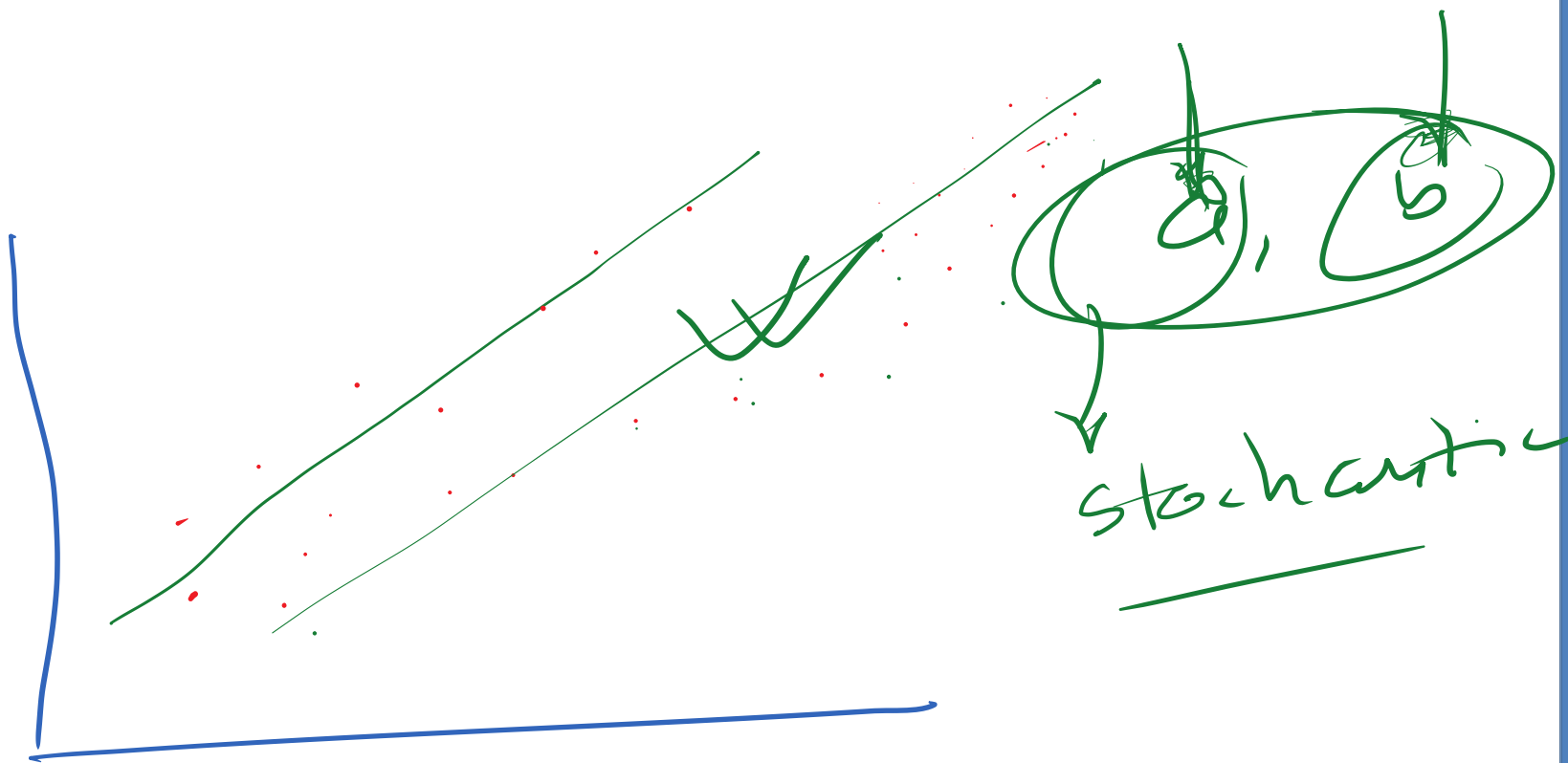
Slower

Faster mover

Accuracy



Speed



$$M+ = w_0 x_0 + w_1 x_1 + \dots + w_n x_n$$

$$M+ = a + b \text{ (leg)}$$

$$g = w^T x \quad \sum w_i x_i$$

ANN

$$g = w_0 x_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

Implications of Fitts' law

- Implications of Fitts' law in HCI design:
 - The slope of the function (b) may vary across different control devices, in which case movement times will be faster for the devices that yield lower slopes.
 - Card, English, and Burr (1978) conducted a study that evaluated how efficient text keys, step keys, a mouse, and a joystick are at a text-selection task in which users selected text by positioning the cursor on the desired area and pressing a button or key.

Memory

- Memory stores factual knowledge and contains knowledge of actions or procedures.
- Memory allows us
 - To repeat actions
 - To use languages
 - To receive new information through our sensors
- Memory gives us the sense of identity, by preserving information from our past experiences.
- Some questions that leads us to know the capability and limitations of memory are:
 - How does memory work?
 - How do we remember arbitrary lists such as those generated in memory games?
 - Why do some people remember more easily than others?
 - What happen when we forget?

Memory...

There are three types of memory function:

Sensory memories



Short-term memory or working memory



Long-term memory

Selection of stimuli governed by level of arousal.

Sensory memory

- The sensory memory act as buffers for stimuli received through the senses.
- A sensory memory exists for each sensory channel:
 - iconic memory: visual stimuli
 - Information remains in iconic memory very briefly, in order of 0.5 seconds.
 - echoic memory: aural stimuli
 - Allows brief 'play-back' of information.
 - haptic memory: tactile stimuli
- Examples
 - “sparkler” trail
 - stereo sound
- Continuously overwritten

Iconic Memory



Short-term memory (STM)

- Short-term memory or working memory act as scratch-pad for temporary recall.
 - E.g. Calculate the multiplication, 45×3
- Short-term memory has the properties of :
 - rapid access $\sim 70\text{ms}$
 - rapid decay $\sim 200\text{ms}$
 - limited capacity - 7 ± 2 chunks
 - What is your digit span?
- Two basic methods of measuring short-term memory:
 - Determining the length of sequence that can be remembered in order
 - Items to be freely recalled in any order

Examples

212348278493202

Examples

11 224 124 6060

HEC ATR ANU PTH ETR EET

Examples

11 224 124 6060

Chunking of information can increase the short-term memory capacity

Examples

HEC ATR ANU PTH ETR EET

Patterns can be useful as aids to memory.

Long-term memory (LTM)

- Repository for all our experimental knowledge, factual information, procedural rules of behaviour.
 - slow access $\sim 1/10$ second
 - slow decay, if any
 - huge or unlimited capacity
- Two types of LTM
 - *Episodic*: represents our memory of events and experiences in a serial form.
 - *Semantic*: is the structured memory of facts, concepts, and skills that we have acquired.
 - The information in semantic memory is derived from that of episodic memory, such that we can learn new facts or concepts from our experiences.

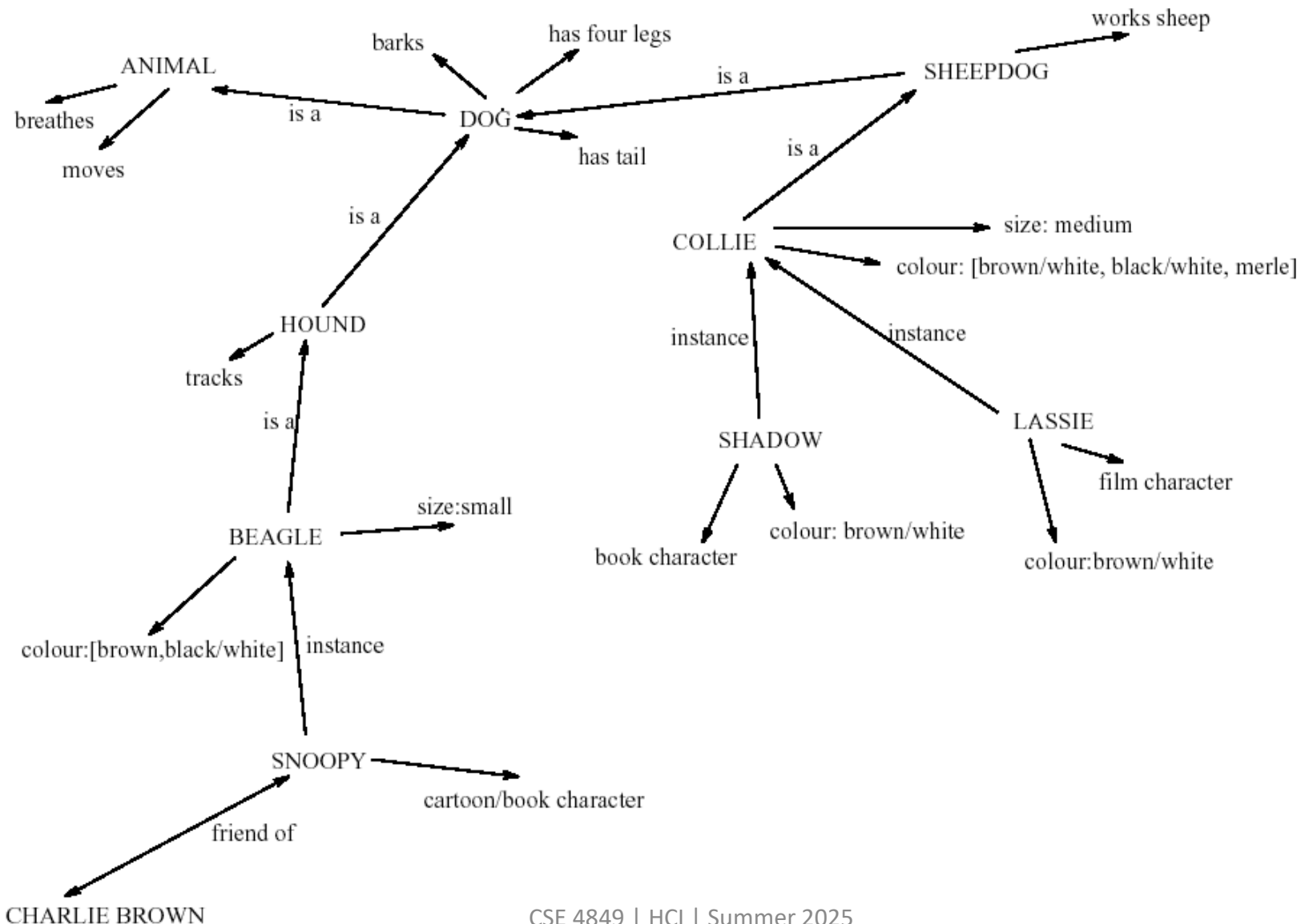
Long-term memory (cont.)

- Semantic memory structure
 - provides access to information
 - represents relationships between bits of information
 - supports inference
- Model: semantic network
 - inheritance – child nodes inherit properties of parent nodes
 - relationships between bits of information explicit
 - supports inference through inheritance

Representations of knowledge

- Semantic network representation
- Frame-based knowledge representation
- Script-based knowledge representation
- Condition-action rules as production rules

LTM - semantic network



Frame-based knowledge representation

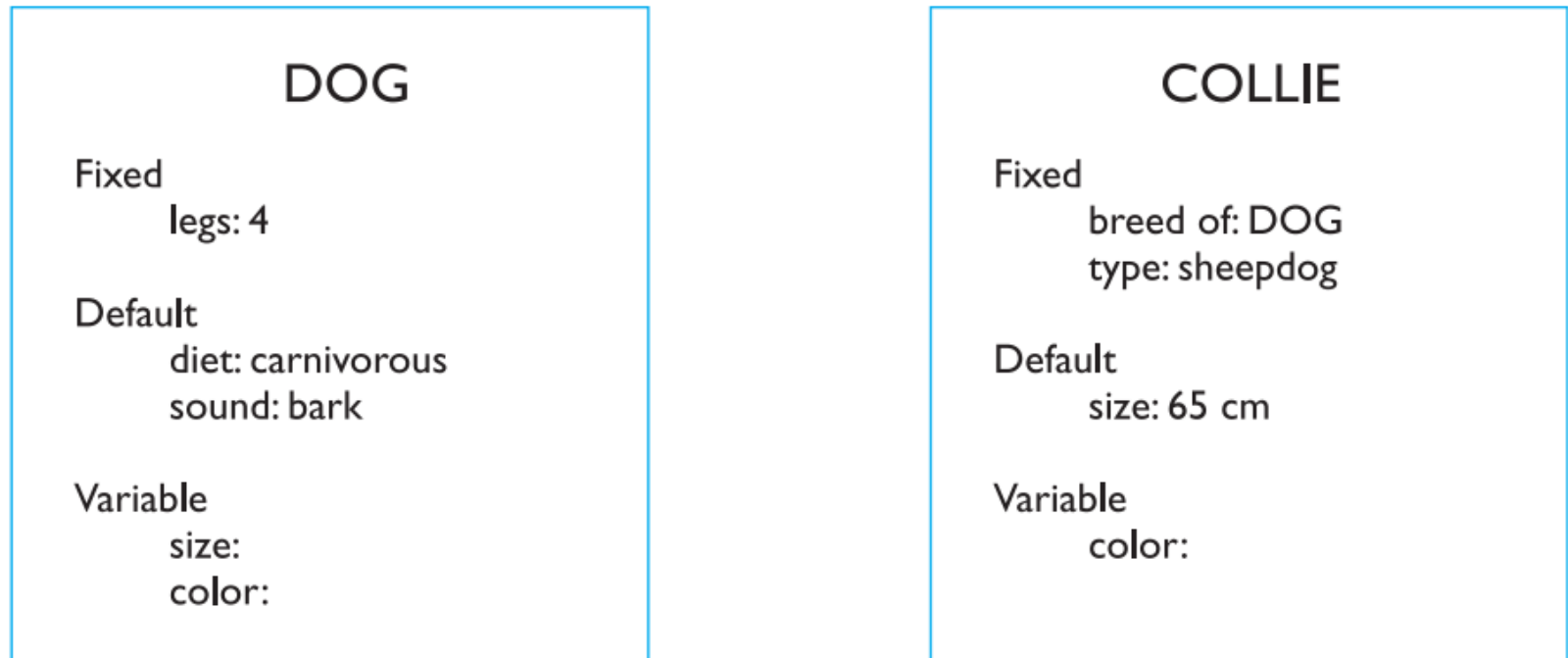


Figure 1.12 A frame-based representation of knowledge

Script-based knowledge representation

Script for a visit to the vet			
Entry conditions:	<i>dog ill</i> <i>vet open</i> <i>owner has money</i>	Roles:	<i>vet examines</i> <i>diagnoses</i> <i>treats</i> <i>owner brings dog in</i> <i>pays</i> <i>takes dog out</i>
Result:	<i>dog better</i> <i>owner poorer</i> <i>vet richer</i>	Scenes:	<i>arriving at reception</i> <i>waiting in room</i> <i>examination</i> <i>paying</i>
Props:	<i>examination table</i> <i>medicine</i> <i>instruments</i>	Tracks:	<i>dog needs medicine</i> <i>dog needs operation</i>

Figure 1.13 A script for visiting the vet

Entry conditions Conditions that must be satisfied for the script to be activated.

Result Conditions that will be true after the script is terminated.

Props Objects involved in the events described in the script.

Roles Actions performed by particular participants.

Scenes The sequences of events that occur.

Condition-action rules-based knowledge representation

- Condition-action rules are stored in long-term memory.
- Information coming into short-term memory can match a condition in one of these rules and result in the action being executed.

IF dog is wagging tail
THEN pat dog

IF dog is growling
THEN run away

LTM processes

- There are three main activities related to long-term memory:
 1. Storage or remember of information
 2. Forgetting the information
 3. Retrieval of information

LTM - Storage of information

- Rehearsal
 - information moves from STM to LTM
- Total time hypothesis [Ebbinghaus]
 - amount retained proportional to rehearsal time
- Distribution of practice effect [Baddeley]
 - optimized by spreading learning over time
- Structure, meaning and familiarity
 - information easier to remember

- Which list is easier to remember?

List A: Faith Age Cold Tenet Quiet Logic Idea Value Past Large

List B: Boat Tree Cat Child Rug Plate Church Gun Flame Head

LTM - Forgetting

Decay

- information is lost gradually but very slowly

Interference

- new information replaces old: retroactive interference
- old may interfere with new: proactive inhibition

so may not forget at all memory is selective ...

... affected by emotion – can subconsciously 'choose' to forget

LTM - retrieval

Recall

- Information reproduced from memory can be assisted by cues, e.G. Categories, imagery

Recognition

- Information gives knowledge that it has been seen before
- Less complex than recall - information is cue

Thinking

Reasoning

deduction, induction, abduction

Problem solving

Reasoning

- Reasoning is the process by which we use the knowledge we have to draw a conclusion or infer something new about the domain of interest.
- Different types of reasoning are:
 - Deductive
 - Inductive
 - Abductive

Deductive Reasoning

- Deduction:
 - Derive logically necessary conclusion from given premises.
e.g. If it is friday then she will go to work
It is friday
Therefore she will go to work.
- Logical conclusion not necessarily true:
e.g. If it is raining then the ground is dry
It is raining
Therefore the ground is dry

Deduction (cont.)

- When truth and logical validity clash ...

e.g. Some people are babies

Some babies cry

Inference - Some people cry

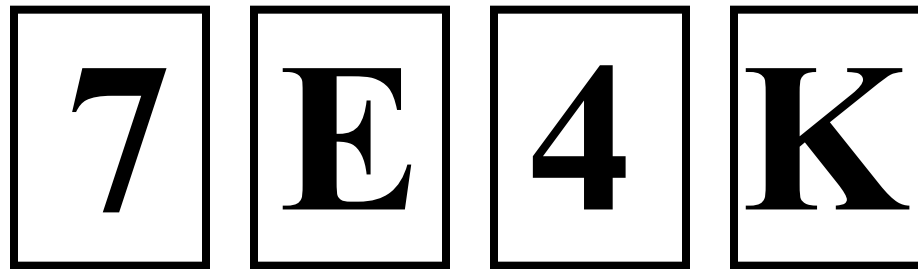
Correct?

- People bring world knowledge to bear

Inductive Reasoning

- Induction:
 - Induction is generalizing from cases we have seen to infer information about cases we have not seen.
 - e.g. all elephants we have seen have trunks therefore all elephants have trunks.
 - Unreliable:
 - can only prove false not true
- ... but useful!
- Humans not good at using negative evidence
 - e.g. Wason's cards.

Wason's cards



If a card has a vowel on one side it has an even number on the other

Is this true?

How many cards do you need to turn over to find out?

.... and which cards?

Abductive reasoning

- Reasoning from event to cause
 - e.g. Sam drives fast when drunk.
 - If I see Sam driving fast, assume drunk.
- Unreliable:
 - Can lead to false explanations

Problem solving

- If reasoning is a means for inferring new information from what is already known, problem solving is the process of finding a solution to an unfamiliar task, using the knowledge we have.

Errors and mental models

Why do we make mistakes and can we avoid them?

In order to answer these question we need to understand the types of error,

- Slips
 - right intention, but failed to do it right
 - Causes: poor physical skill, inattention etc.
 - Change to aspect of skilled behaviour can cause slip
- Mistakes
 - Wrong intention
 - Cause: incorrect understanding

Humans create mental models to explain behaviour.
If wrong (different from actual system) errors can occur

Affective Computing Emotion

Human Affection

- Various theories of how emotion works
 - James-Lange: emotion is our interpretation of a physiological response to a stimuli
 - Cannon: emotion is a psychological response to a stimuli
 - Schacter-Singer: emotion is the result of our evaluation of our physiological responses, in the light of the whole situation we are in
- Emotion clearly involves both cognitive and physical responses to stimuli

Trust
Sati

Emotion (cont.)

- The biological response to physical stimuli is called *affect*
- Affect influences how we respond to situations
 - positive → creative problem solving
 - negative → narrow thinking

“Negative affect can make it harder to do even easy tasks; positive affect can make it easier to do difficult tasks”

(Donald Norman)

Emotion (cont.)

- Implications for interface design

- Stress will increase the difficulty of problem solving
- Relaxed users will be more forgiving of shortcomings in design
- Aesthetically pleasing and rewarding interfaces will increase positive affect

Individual differences

- Long term
 - Sex, physical and intellectual abilities
- Short term
 - Effect of stress or fatigue
- Changing
 - Age

Ask yourself:

Will design decision exclude section of user population?